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Laser excitation of high-frequency capillary waves

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Abstract

The laser-induced surface deformation (LISD) technique was applied to generate high-frequency capillary waves on liquid surfaces up to several tens of kHz in a noncontact manner. The dynamic response of the fluid near the surface was theoretically derived under the condition of periodical radiation pressure. The result of the numerical calculation predicts the propagation of induced capillary waves out from the excitation region. The efficiency of the wave generation was experimentally examined by changing the width of the excitation laser beam at the surface. The observed LISD spectra were well reproduced by the theory, showing that the effective frequency band can be extended up to over 100 kHz. The propagation of the optically generated wave was measured with a laser probe sweeping the position of the observation. The spatial profile gives the surface tension and the shear viscosity of the sample liquid. The frequency domain measurement was also carried out and the spectrum obtained at a fixed point agrees with the theory, demonstrating the rapid measurement of frequency-dependent phenomena.

Keywords: Capillary wave; Optical radiation pressure; Laser-induced surface deformation; Ripplon spectroscopy; Surface tension; Wave excitation

1. Introduction

Spectroscopic measurement of capillary wave propagation is an effective tool to characterize the dynamic properties of liquid surfaces and liquid–liquid interfaces. The frequency dependence of the surface properties derived from the dispersion of the capillary wave gives information on the dynamic behavior of the molecules on and near the surface [1–3]. The surface pressure and surface viscoelasticity are, in particular, good measures to characterize the state of two-dimensional condensation in the molecular layer adsorbed on the surfaces.

Two major techniques, ripplon light scattering [4,5] and excited capillary wave measurements [1,2,6], have been widely employed in various fields of liquid surface science. While the observation of the ripplon, a thermally excited capillary wave, has the advantages of noncontact and high-frequency measurements, the observation of a coherent wave artificially excited with well-defined phase brings high accuracy. The accuracy of the measurement in a recent experiment was improved to 10 μN/m for the relative change in the surface tension. Such high accuracy is quite important in investigating the dilute molecular layer on the surface. Various kinds of surface wave excitation methods have been proposed so far besides the conventional mechanical drive of the surface oscillation. Sohl et al. have excited surface waves by dielectric interaction between an electrode blade and a liquid sample [7]. The dynamic surface displacement due to the wave propagation is detected by laser diffraction. Though this method is completely noncontact, the excitation frequency is limited since the shortest wavelength is roughly determined by the spacing between the blade and the surface. Focused ultrasonic waves have also been used as a wave source, which is useful only for capillary waves with wavelength shorter than the ultrasonic wavelength as easily understood [8]. We have proposed mechanical excitation with an electromagnetic wire transducer, in which a very thin metal wire conducting alternating current moves up and down in a horizontal magnetic field and generates a capillary wave [6]. The wavelength is restricted roughly by the wire thickness and the frequency band is extended up to several tens of kHz. In each method, the frequency range is limited below the kHz range due to the finite characteristic size of the excitation device.

In order to avoid these restrictions common to conventional measurements, we introduce a novel method of exciting the capillary waves by applying the periodical radiation pressure of laser light to a surface. The newly developed system of laser-induced surface deformation (LISD) demon-

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strated a high potential for noncontact and contamination-free measurement, as well as wide frequency observation.

2. Excitation of liquid surface waves by periodical laser radiation

The radiation pressure of a laser beam applied to the interface between two media with different indices of refraction brings about the deformation of the liquid interface. Another probe laser detects the induced deformation, whose amplitude is typically of nanometer order. We can obtain the local mechanical properties of the liquid surface by measuring the induced surface curvature [9,10]. In addition, the periodical modulation of the laser radiation also excites the dynamic motion of the surface, which yields dynamic information on the molecular physics at the liquid surface. The LISD technique can induce very local surface deformation and get information for an area as small as 10 µm. The accuracy of the surface properties are restricted due to the limited area of observation, however, since the determination of the beam profile at µm size is quite difficult, for example.

More accurate measurements would be possible if the deformation propagated as a wave for a distance and its velocity and damping constant were detected at a point apart from the position of excitation. We considered that the LISD technique would be also useful as a tool to generate coherent capillary waves.

The principle for the optical deformation of the liquid surface has previously been described in detail [9] and we give here a brief account. The propagating light carries momentum, which is given by $p = I_0/c$, where I_0 is the intensity of the light and c is the light velocity. The momentum of the light changes discontinuously when it penetrates the interface separating two media with different light velocities. This difference in the momentum works as radiation pressure and the surface is deformed so that the Laplace force of the curved interface balances the external pressure. The periodical modulation of the laser intensity induces a coherent wave at the required frequency.

The optical generation of the liquid surface wave has several remarkable advantages over conventional mechanical contact transducers. First, in the noncontact method, the remote sensing and processing of the liquid surface are possible from locations apart from the sample. The technique can therefore be applied to experiments under the extraordinary environments such as high or ultra-low pressures and high temperatures. It can easily be adopted also for the investigation of liquid/liquid interface. It has also the advantage of being a nondestructive approach to a surface with structures of molecular layers.

Second, we anticipate the extension of the measurement frequency band up to a higher frequency region, which has been limited in principle by the spatial dimension of the excitation device or, in some cases, by the length of the curved meniscus of the surface in touch with the transducer. The maximum frequency of the artificial capillary wave has so far been about 40 kHz for a pure water surface. Optical excitation has the potential to generate higher frequency waves by focusing the laser beam to the optical diffraction limit. The frequency range can be expanded to several MHz, far beyond the present limit.

The technique has another advantage: it could induce various modes of surface vibration, such as plane and circular surface waves, by focusing the exciting laser beam in an appropriate shape. Line focusing by a cylindrical lens would generate a plane wavefront propagating in a unique direction.

The greatest advantage is that we can quantitatively estimate the waveform of the excited surface wave, including the amplitude and phase, since the spatial distribution of the radiation pressure is easily available by observing the laser beam profile at the liquid surface. As described later in detail, the amplitude and phase of the excited surface wave are theoretically given at an arbitrary frequency, which enables us to measure the surface wave propagation in the frequency domain at a fixed observation point.

Here, we theoretically derive the behavior of the liquid under an external surface force. We set an orthogonal coordinate so that the surface is on the x–y plane with the liquid filled in $z < 0$ and the wave with linear wavefront propagates into the $+x$ direction. The focus of the pump laser beam is centered at the origin of the x–y plane. We made an approximation that the focus of the cylindrical lens is long enough in the y-direction and we only consider the motion of the liquid in the x- and z-directions. The fluid motion is given by solving the Navier–Stokes equation for an incompressible fluid,

$$\frac{\partial \boldsymbol{v}}{\partial t} = -\frac{1}{\rho}\operatorname{grad} p + \frac{\eta}{\rho}\Delta \boldsymbol{v}, \quad \operatorname{div} \boldsymbol{v} = 0, \tag{1}$$

under the boundary condition that the hydrodynamic force is balanced with the sum of the Laplace force of the curved surface and the external radiation pressure, which is represented by

$$-p + 2\eta \frac{\partial v_z}{\partial z} = \sigma \frac{\partial^2 \xi}{\partial x^2} + f_e(x,t). \tag{2}$$

Here, p is the pressure, η the viscosity of the substrate liquid, σ the surface tension, $\boldsymbol{v}$ the velocity vector and v_z its z-component, ξ the surface displacement, and f_e the external force applied to the unit surface area. We decompose the periodical external surface force into the spatial Fourier components represented by the wavenumber k as

$$f_e(x,t) = e^{i\omega t} \int f_e(k) e^{ikx}\, dk. \tag{3}$$

By substituting the boundary condition of Eq. (2) into Eqs. (1) and (3), we obtain the surface displacement with wavenumber k as

$$\xi_k = \frac{k}{\gamma^2 \rho} D(u)^{-1} f_e(k), \tag{4}$$

where

$$D(u) = (u+1)^2 + (\omega_0/\gamma)^2 + \sqrt{1-2u}$$
$$(\mathrm{Re}[\sqrt{1-2u}] > 0),$$
$$\omega_0 = (\sigma/\rho)^{1/2} k^{3/2},$$
$$\gamma = 2(\eta/\rho)k^2,$$
$$u = i\omega/\gamma.$$

The final form of the surface displacement is then given by the integration of Eq. (4) with an appropriate function of $f_e(k)$. As for the Gaussian beam profile of $f_e(x,t) = f_0 \exp(-i\omega t)\exp(-2x^2/w^2)$, the Fourier component is given by $f_e(k) = f_0 \exp(-k^2 w^2/8)$ and the displacement is then obtained as

$$\xi(x,t) = f_0 e^{-i\omega t} \int \frac{k}{\gamma^2 \rho} D(u)^{-1} \exp(-k^2 w^2/8) e^{ikx}\, dk. \tag{5}$$

We neglected the effect of the gravity in the above discussion, since it only causes slight deformation of the surface due to the meniscus ranging up to as large as ∼mm.

Figure 1 is the result of the numerical calculation of Eq. (5) showing the periodical behavior of the surface under the radiation sinusoidally modulated at $\omega/2\pi = 1$, 10, 30, and 100 kHz in Figs. 1a, 1b, 1c, and 1d, respectively. The beam profile is a Gaussian with width $w = 70$ µm, which is also shown schematically in the figure.

As shown in the figure, the surface deformation induced at around $x = 0$ propagates outward when wavelengths are comparable to or larger than the characteristic size of the induced deformation. The waves show spatial damping as they propagate. The dashed line in (c) represents exponential decay with the literature value of the spatial damping constant $\alpha = 3.1 \times 10^3\ \mathrm{m}^{-1}$, which well reproduces the envelope of the calculated curves in the region $x > w$. The spatial decay should be measured in the area out of the beam width. On the other hand, the high-frequency waves with shorter wavelength cannot propagate even though surface deformation is slightly induced near the radiation area at the given frequency. It is interesting to point out that the amplitude of the propagating wave increases and becomes larger than the induced surface displacement as apparently shown in (b) and (c). The laser radiation pressure does not essentially induce the deformation but applies a force to the surface. The amplitude would increase as the wave propagates. This situation is analogous to the fact that we can cause a large vibration of a stretched rope with a small swing of our hands.

3. Experiment

The block diagram of the experimental setup is schematically shown in Fig. 2. A frequency-doubled cw-YAG laser with a maximum output power of 5 W is used as a light source for wave generation. The laser is once conducted

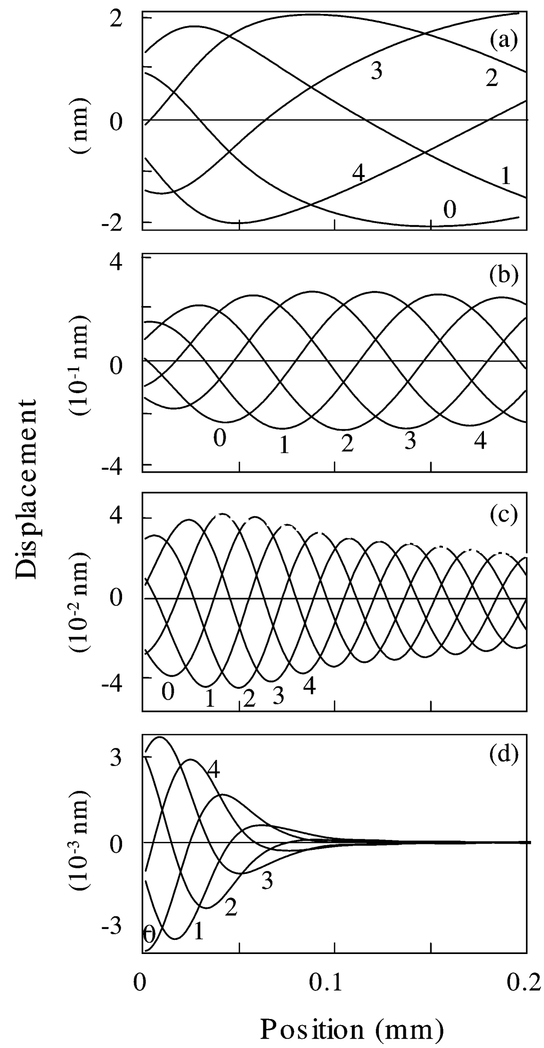

Fig. 1. Numerically calculated surface deformation under the periodical radiation pressure obtained at (a) 1 kHz, (b) 10 kHz, (c) 30 kHz, and (d) 100 kHz. The numbers show the temporal evolution of the surface form as (0) $t = 0$, (1) $t = (1/5)T$, (2) $t = (2/5)T$, (3) $t = (3/5)T$, and (4) $t = (4/5)T$, where T is the vibration period given by $T = 2\pi/\omega$.

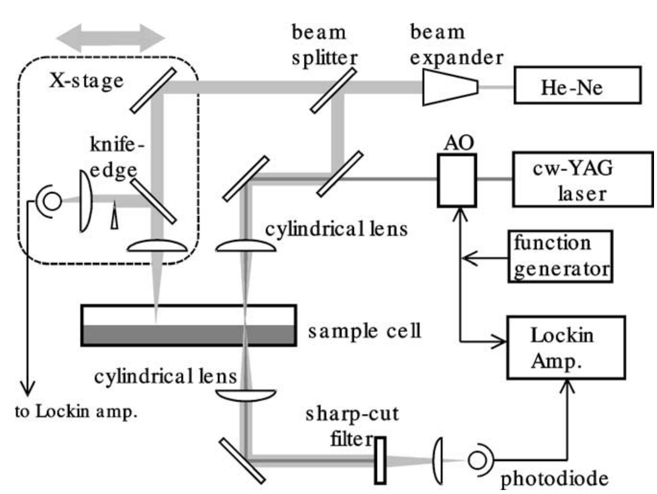

Fig. 2. Block diagram of the experimental setup.

through an acousto–optical (A–O) modulator and the first-order diffracted light is incident to the liquid surface. The intensity of the incident laser can be modulated to an arbitrary temporal profile by controlling the driving power of the A–O modulator in the frequency band from DC to 1 MHz.

The incident laser beam is then focused into a narrow elliptic profile on the liquid surface by a cylindrical lens. The intensity distribution is ideal Gaussians in both the x and y directions. The length is about 1 mm in the y direction, which is taken to be larger than the typical wavelength of the excited surface wave at kHz region. The width, on the other hand, is changed by choosing appropriate focal lengths of lenses. The beam width w is roughly related to the focal length f, the wavelength of light λ, and the initial beam diameter D to $w \sim f\lambda/D$. As shown in the later section, the beam width sensitively determines the effective frequency band of the wave generation.

The relation between the surface displacement and the optical intensity of the incident laser has already been discussed. A typical displacement is about 1 nm for an intensity of 1 W focused within an ellipse 1 mm in length and 10 µm in width.

A laser probe detects such a small surface deformation. Two different types of detection method were employed. One is the measurement of the surface curvature just at the excitation point, and the other is the observation of the surface gradient due to the wave propagation by the laser deflection. The former was carried out to estimate the excitation efficiency at the origin excluding the effect of the propagation characteristics of the surface. The latter was the major measurement to observe the wave propagation, which directly leads to the accurate characterizations of the liquid surface properties.

The probe laser is a 10-mW He–Ne laser and the output beam is expanded by a beam expander to diameter 10 mm and then focused. The width of the probe laser at the surface is, therefore, small enough compared with that of the pump laser, whose initial diameter is about 1 mm. The expanded probe laser is divided by a beam splitter and a half portion is superimposed by a dichroic mirror to the pump laser beam. In order to observe the surface curvature, the pump and probe laser beams are focused by the same cylindrical lens onto the same area of the surface. The probe laser penetrates the deformed surface, which works as a cylindrical convex lens. The beam profile of the probe laser is distorted through the transmission, and the change in the beam shape is converted to a change in the intensity by a pinhole set before the photodetector.

To measure the spatial profile of the propagating capillary waves, the periodical deformation of the liquid surface is picked up by observing the deflection of the probe laser reflected at the surface. The incline of the surface due to the wave propagation deflects the probe laser and the angle of reflection is converted to a change in the intensity by a knife-edge set in front of the photodetector. The probe laser is focused by a lens with focal length 50 mm on a spot with diameter as small as 2.5 µm, so we can pick up very small wavelengths. The highest detectable wavenumber is $k \sim 2\pi/\lambda \sim 2 \times 10^6\ \mathrm{m}^{-1}$, which corresponds to the frequency band up to 4 MHz. The expected angle of deflection is roughly estimated as $\xi/\lambda \sim 10^{-6}$ rad with $\xi \sim 1$ nm and $\lambda \sim 1$ mm, with the typical values for the generated waves at 1 kHz, which can be detected with sufficient sensitivity and stability by a combination of the optical lever technique and lock-in detection. The deflection angle is almost constant up to several tens of kHz, since the amplitude of the induced wave is roughly proportional to the inverse of the wavenumber, as shown in Fig. 1.

The position of the probing spot is swept with a linear mechanical stage along the direction of the wave propagation to observe the spatial profile of the capillary wave. The detected signal is analyzed by a lock-in amplifier that works with the laser radiation signal as the reference.

The sample is pure distilled water contained in a teflon-coated trough 100×200 mm in size. The temperature of the sample is kept at $25 \pm 1\,^\circ\mathrm{C}$ throughout the experiment.

4. Results and discussion

4.1. Estimation of effective frequency band

For the purpose of examining the relation between the focusing width of the pump laser and the excitation frequency band, the response functions of the liquid surface at the origin ($x = 0$) are measured by the LISD method for three cylindrical lenses with different focal lengths of 100, 200, and 300 mm. The actual beam width for each lens was measured by a laser beam profiler as $w = 40$, 80, and 120 µm, respectively. Note here that the LISD signal is proportional to the surface curvature, which is given by the second-order differential of the surface displacement function given by Eq. (5). Ignoring the time-dependent factor $\exp(-i\omega t)$ and with $x = 0$, we obtain

$$s(\omega) \propto \int \frac{k^3}{\gamma^2\rho} D(u)^{-1} \exp(-k^2w^2/8)\, dk. \tag{6}$$

Figure 3 shows a typical LISD spectrum observed with a beamwidth of $w = 80$ µm, in which the real part of the signal decreases with frequency and crosses the line $\mathrm{Re}[s(\omega)] = 0$ at a characteristic frequency f_c. This characteristic frequency (shown by the arrow in the figure) is used as the measure to estimate the effective frequency band.

Figure 4 shows the relation between the characteristic frequency and the beam widths. The solid line shows the theoretical prediction derived form the numerical calculation of the response function of Eq. (6). The experimental results agree well with the prediction as shown in the figure, which demonstrates the validity of the theoretical discussion given above.

The theoretical curve is approximately expressed by $f_c \sim (\sigma/\rho)^{1/2} w^{-3/2}$ and we can expect a very wide frequency

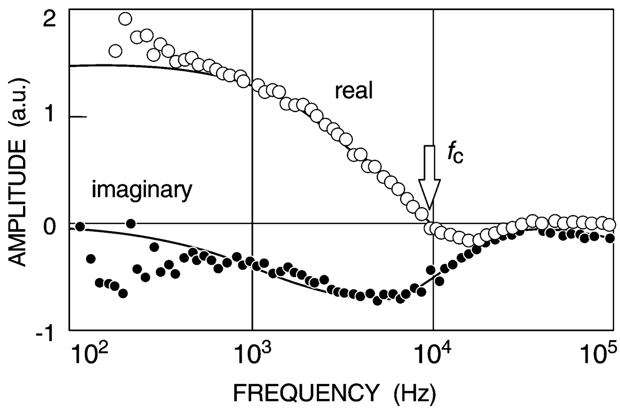

Fig. 3. Typical example of the LISD spectrum obtained with a beamwidth of $w = 80$ μm. The open and closed circles indicate the real and the imaginary part of the complex amplitude, respectively.

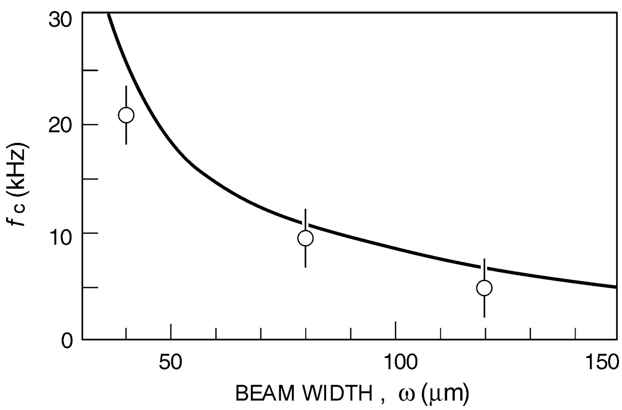

Fig. 4. Dependence of the characteristic frequency on the beamwidth. The solid line represents the theoretical prediction of Eq. (6).

band, up to or even over the MHz region, if the beamwidth approaches the optical diffraction limit: $f_c = 2.4$ MHz for $w = 0.5$ μm, for example. The excitation efficiency, including the amplitude and phase of the induced surface waves at each frequency, is theoretically given with the knowledge of the beam profile at the liquid surface. As described later, the frequency domain measurement of the surface properties is thus possible with the frequency dependence of the excitation efficiency.

4.2. *Measurement of spatial profile at a constant frequency*

The spatial profile of the excited surface wave was measured by sweeping the surface with the probe laser along the direction of the wave propagation. The real and imaginary parts of the detected signal obtained at $f = 5$ kHz are shown as functions of the position in Fig. 5a. From the phase shift shown in (a) and the amplitude (b), we obtain the phase velocity v_p and the spatial damping constant α, which lead to the surface tension and the shear viscosity of water, respectively, through the relation of $v_p = (\sigma\omega/\rho)^{1/3}$ and $\alpha = 4\eta\omega/3\sigma$. The solid lines show the theoretical curve drawn with the literature values of $\sigma = 7.196 \times 10^{-2}$ N/m, $\rho = 998$ kg/m^3, and $\eta = 0.89$ N s/m^2. The experimental result shows remarkable agreement with the theory.

Here, we have to take the effect of the wave diffraction into account, since the lateral length of the focused beam is 1 mm, almost comparable to the wavelength at kHz region. The angler broadening of diffraction is roughly given by λ/l, λ, and l being the wavelength and the length of the focus, respectively, and the wavefront broadens from l to $l + L\lambda/l$ after propagation distance L. By neglecting the absorption of the capillary wave, the amplitude of the wave decreases to $(1 + L\lambda/l^2)^{-1/2}$. As for the frequency of 5 kHz, the wavelength is 0.26 mm on pure water and the decrease in amplitude is estimated to be about 9% at $L = 1$ mm. The decay due to the intrinsic absorption is about 40% at this frequency, which is larger than the effect of the diffraction. The influence of the diffraction is not so serious at high frequencies with an appropriate correction. However, we have to pay attention when we apply the system to a lower frequency region.

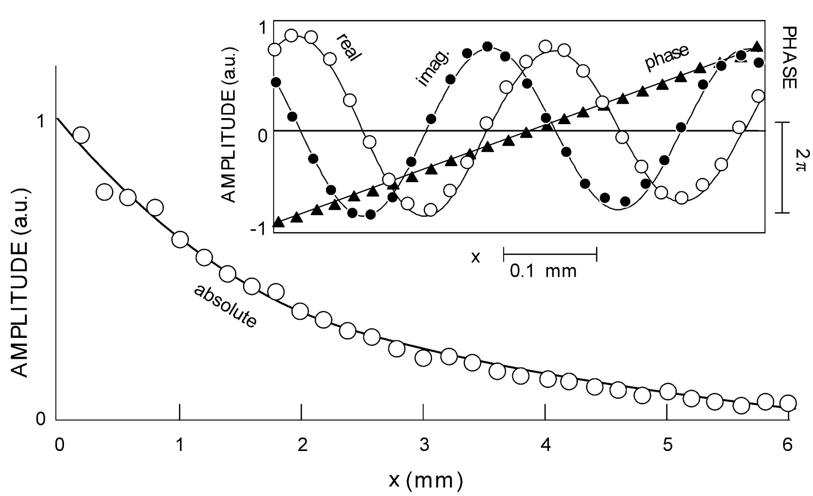

Fig. 5. Exponential decay of the laser-excited capillary waves observed at 5 kHz. The inset figure shows the real part (open circles), the imaginary part (closed circles), and the phase (triangle) of the complex amplitude on an expanded abscissa. The solid lines are the corresponding theoretical curves.

4.3. Frequency domain observation of capillary wave propagation

As described previously, the optical generation has the remarkable advantage that the frequency dependence of the excited wave amplitude is theoretically calculated. This enables a new method of surface wave measurement in the frequency domain. As clearly shown in Fig. 1, the oscillation at the origin is not necessarily in phase with that of the pump laser. The shape of the surface is complicated around $x = 0$ and the origin does not necessarily agree with the origin of phase of the propagating wave. In addition, the amplitude of the excited capillary wave depends on the frequency and the beam width as described previously. We have to extract the influence of these factors dependent on the experimental conditions. This effect is represented by the complex amplitude $A^*(\omega)$ and the signal observed is described as

$$S(\omega, x) \propto \frac{\partial \xi}{\partial x} = A^*(\omega)(ik_0 - \alpha)e^{-i\omega t} \exp\{(ik_0 - \alpha)x\}. \tag{7}$$

Here, the wavenumber k_0 is given by $k_0 = (\rho\omega^2/\sigma)^{1/3}$. The prefactor $A^*(\omega)$ can be obtained either by the numerical calculation of Eq. (5) or by the actual measurement of the frequency dependence of the wave amplitude and relative phase near the excitation point.

If the surface tension and the viscosity are independent of the frequency, surface properties are obtained by fitting Eq. (7) to the experimental data. The experimental results actually obtained with the beamwidth of $w = 40$ µm and at a few points of x are shown in Fig. 6. The solid line shows the theoretical curve drawn with the literature values of the surface properties, which is in very good agreement with the experimental result. The frequency domain measurement has another advantage, that the system is free from the mechanical noise.

In conclusion, we successfully applied the laser-induced surface deformation technique to the generation of high-frequency capillary waves. We could measure the surface wave propagation at a fixed observation point in the frequency domain, which enables us to observe the real-time measurement of the surface dynamic prosperities. Frequency dependence of the propagation characteristics brings important information on the surface viscoelasticity of the molecular covered surface. The laser excitation method has a few more advantages: we can generate an arbitrary waveform by modulating the laser output. It is also possible to excite the interfacial waves between the liquids completely in a non-contact manner. The experiments of the oil/surfactant/water system are already started and the results would soon be reported.

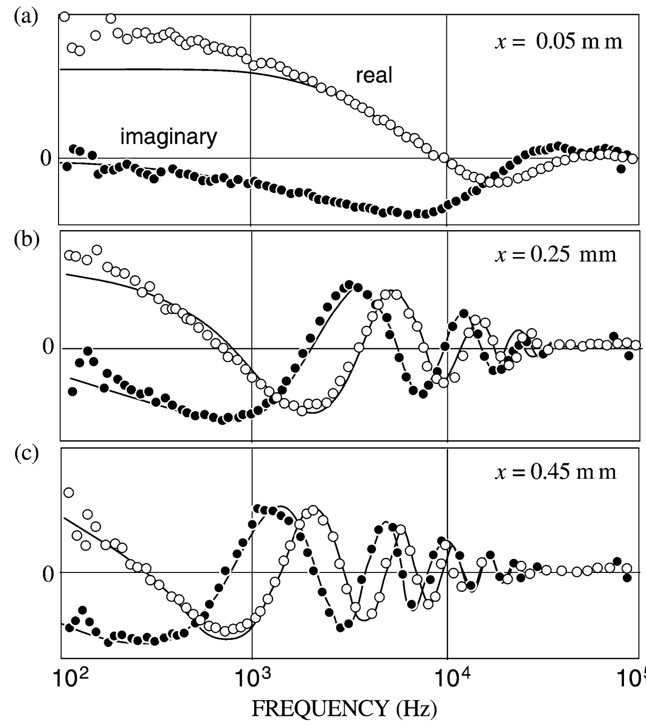

Fig. 6. Frequency spectrum of the propagating capillary waves observed at $x =$ (a) 0.05 mm, (b) 0.25 mm, and (c) 0.45 mm. The solid lines are the fitted curves of Eq. (7).

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